JEE

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Maths

Trigonometric Equations

DPP: 4

Q1 $\sin \theta+\sin 5 \theta=\sin 3 \theta$, where $0 \leq \theta \leq \pi$.

Q2 $4 \sin x \cdot \sin 2 x \cdot \sin 4 x=\sin 3 x$.

Q3 $\sin ^6 2 x+\cos ^6 2 x=\frac{7}{16}$.

Q4 $\sin x+\sin 2 x+\sin 3 x=\frac{1}{2} cot \frac{x}{2}$.

Q5 $5 \sin ^2 x-7 \sin x \ \cos x+16 \cos ^2 x=4$.

Q6 $\sin ^2 \theta-\cos \theta=\frac{1}{4}$, where $0 \leq \theta \leq 2 \pi$.

Q7 $\sin ^4 x-\cos ^7 x=1$.

Q8 $\sin \frac{2 x+1}{x}+\sin \frac{2 x+1}{3 x}-3 \cos ^2 \frac{2 x+1}{3 x}=0$.

Q9 In a right angled $\triangle ABC$, the hypotenuse is four times the perpendicular distance of the opposite vertex from it. Determine the other two angles.

Q10 Solve: $\cos 3 x \cos ^3 x+\sin 3 x \sin ^3 x=0$.

Q11 Solve: $\sin 3 \theta-\sin \theta=4 \cos ^2 \theta-2$.

Q12 Solve: $sec 4 \theta-sec 2 \theta=2$.

Q13 Solve: $\sin ^2 n \theta-\sin ^2(n-1) \theta=\sin ^2 \theta$.

Q14 Solve: $2^{\cos 2 x}=3.2^{\cos ^2 x}-4$.

Q15 Solve: $\cos \frac{\sqrt{2}+1}{2} \theta \cos \frac{\sqrt{2}-1}{2} \theta=1$

Q16 Solve: $\tan \left(\frac{\pi}{4}+x\right)+\tan \left(\frac{\pi}{4}-x\right)=2$

Answer Key

Q1 $0, \frac{\pi}{6}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{5\pi}{6} \ \& \ \pi$

Q2 $x = n\pi \ or \ \frac{m\pi}{3} \pm \frac{\pi}{9}, \ n, m \in I$

Q3 $x = \frac{n\pi}{4} \pm \frac{\pi}{12}, \ n \in I$

Q4 $x = \frac{2n\pi}{7} + \frac{\pi}{7}, \ n \in I$

Q5 x = $n\pi + \tan^{-1} 3$ or $m\pi + \tan^{-1} 4, \ m, \ n \in I$

Q6 $\theta = \frac{\pi}{3}, \ \frac{5\pi}{3}$

Q7 $x = 2n\pi + \pi, \ n \in I, \ x = n\pi + \frac{\pi}{2}, \ n \in I$

Q8 $x = \frac{2}{6n\pi+3\pi-4}$ or $\frac{1}{3n\pi+3(-1)^n \sin^{-1} \frac{3}{4} - 2}$ where $n \in I$

Q9 15°, 75°

Q10 $x = |2n+1| \frac{\pi}{4}, \ n \in I$

Q11 $\theta = \left(2n+1\right) \frac{\pi}{4} \ or \ \theta = 2n\pi + \frac{\pi}{2}$

Q12 $\theta = \left(2n+1\right) \frac{\pi}{2}$ or $\theta = \left(2m+1\right) \frac{\pi}{10}, \ m, \ n \in I$

Q13 $\theta = m\pi, \ m \in I \ or, \ \theta = \frac{m\pi}{n-1}, \ m \in I \ or, \ \theta = \left(2m+1\right) \frac{\pi}{2n}, \ m \in I$

Q14 $x = n\pi, \ n \in I$

Q15 0

Q16 x=$n\pi, \ n \in I$